ROUND 2 & MVP / WORKING PRODUCT PRESENTATION

TRVST AI

Intelligent Brand Governance Platform

A practical review layer that helps teams publish on-brand, accurate, accessible content with people in control.

01. GUIDELINES VAULT → 02. DUAL AI REVIEW → 03. HUMAN APPROVAL LOOP

ENGINEERING TEAM

Team Spider

Full-Stack Architecture & Working Product Dossier

OCTOBER 2026

LIVE ON GITHUB PAGES & VERCEL

01 / THE CHALLENGE

Consistency Gets Harder at Scale

Publishing across omnichannel networks multiplies the number of contributors, assets, and regulatory facts that must stay aligned.

Voice Drifts

TONE

Different writers, agencies, and regional teams interpret voice differently, diluting enterprise brand authority.

Facts Go Stale

PRICING

Pricing tiers, SLA guarantees, and security certifications change frequently, leading to conflicting customer-facing claims.

Claims Carry Risk

LEGAL LIABILITY

Unqualified promises like "100% risk-free" or missing safe-harbor disclaimers expose companies to regulatory penalties.

Visual Standards Missed

BRAND IDENTITY

Logo margins, skewing, and unauthorized color codes degrade visual brand integrity across social and ad channels.

Access Overlooked

ACCESSIBILITY

Sub-4.5:1 color contrast and missing descriptive alt-text violate WCAG AA requirements and exclude user segments.

Review Gets Stuck

VELOCITY

Ambiguous, late feedback loops create painful revision cycles, delaying go-to-market campaigns by days or weeks.

The Product Opportunity: Turn dormant 50-page brand books into an active, automated pre-publication review checkpoint directly inside the creator workflow.

02 / TARGET PERSONAS

One Unified Review Across Four Teams

Engineered to balance creative agility with regulatory certainty through role-tailored governance workflows.

✍️ Creators & Designers

Needs: Fast Feedback & Low-Friction Fixes

Receives instant pre-publication screening with 1-click compliant rewrites, ensuring creative velocity isn't hindered by bureaucracy.

🎯 Brand & Marketing Leads

Needs: Voice Consistency & Exception Tracking

Maintains unified standards across external agencies and internal squads. Identifies repeat rule confusions via real-time analytics.

🏢 Agencies & Freelancers

Needs: Instant Alignment Without Reading 50-Page PDFs

Pre-tests copy against approved corporate facts and terminology before handoff, eliminating repetitive revision cycles.

⚖️ Legal & Compliance Officers

Needs: Evidence Citations, Severity Gating & Sign-Off Control

Critical legal risks are automatically gated and escalated. An immutable audit trail logs every approval, override, and qualification.

Shared Success Condition: A creator understands exactly what to change in 10 seconds; an approver sees why it matters and retains final decision authority.

03 / PRODUCT PHILOSOPHY

A Review Partner, Not a Black Box

TRVST AI combines automated multimodal checks with transparent human judgment so publishing authority remains with people.

1

CHECK

Compares copy and visual assets against approved Knowledge Vault rules, facts, terms, and WCAG standards in real-time.

2

EXPLAIN

Shows the flagged excerpt, exact rule citation, clear rationale, and an editable 1-click compliant rewrite preserving intent.

3

DECIDE

Empowers human reviewers to accept rewrites, dismiss stylistic preferences, escalate to legal, or grant formal publishing approval.

Explainable
Every flag links directly to a verifiable rule or corporate source.

Helpful
Rewrites preserve context and show exact diffs.

Proportionate
Separates critical legal blockers from minor style preferences.

Human-Led
AI advises and guides; humans make the final publication call.

Interactive Split-Screen Review Workspace

Live in production: A unified single-page application connecting content ingestion with explainable findings and one-click fixes.

01. Setup Vault → 02. Submit Content → 03. Dual AI Review → 04. Understand Flags → 05. 1-Click Fix & Approve → 06. Governance Insights

Left Surface: Ingestion & Live Editor

CREATOR AREA

- **Channel Routing:** Tailors checks for Social Media, Landing Pages, Ad Banners, Blog Posts, and Press Releases.
- **Real-Time Analytics:** Live word count and character count monitors.
- **1-Click Presets:** Preloaded benchmark scenarios allowing instant testing by judges without manual typing.
- **Dynamic Text Replacement:** Automatically updates when suggested rewrites are accepted in the right pane.

Right Surface: Explainable Findings & Action Layer

DECISION AREA

- **Animated Score Dial:** 0-100 Brand Alignment gauge reflecting weighted critical, moderate, and minor violations.
- **Severity Filtering:** Filter cards by Critical Risks, Moderate Inconsistencies, or Minor Style Polishing.
- **Rule Citations:** Explicit reference to Knowledge Vault policies (`BR-01`, `BR-02`, etc.) with regulatory rationale.
- **1-Click Compliant Rewrites:** Instant drop-in substitution preserving original creative intent.

05 / CORE INTELLIGENCE LAYER

Dual-Mode AI Architecture

Engineered with high resilience: combining deep multimodal LLM intelligence with an offline, zero-downtime semantic heuristic engine.

Mode A: Google Gemini 2.5 Flash

LIVE AI STUDIO API

Direct REST connection to Google AI Studio for deep contextual and semantic evaluation.

- **Strict JSON Schema Enforcement:** Forces structured output adhering to our typed review model.
- **Dynamic In-App Key Configuration:** Users can enter custom API keys via the UI settings drawer without server restarts.
- **Contextual Rewriting:** Leverages deep semantic understanding to generate nuanced, enterprise-grade revisions.
- **High-Speed Low-Latency:** Sub-second response times on standard broadband connections.

Mode B: TRVST Semantic Heuristic Engine

OFFLINE ZERO-FAILURE GUARANTEE

Embedded deterministic pattern & rule engine operating client-side in pure JavaScript.

- **Zero Cloud Dependencies:** Runs even with complete internet loss or rate limit exhaustion.
- **Heuristic Vector Matching:** Scans for banned terms, retired $29/mo rates, unverified SOC1 credentials, and puffery.
- **Instantaneous Execution:** Renders complete assessment with severity classifications in under 5 milliseconds.
- **Pitch Protection:** Guarantees hackathon judges will NEVER encounter a crashed demo or blank screen.

Fail-Safe State Machine: The engine automatically attempts Mode A first; if an API key is absent, throttled, or times out, it seamlessly falls back to Mode B without disrupting the user experience.

Codifying Guidelines into Machine-Readable Rules

The Knowledge Vault centralizes brand books, legal disclosures, verified product claims, and accessibility requirements.

Brand Rules & Policies

6 ACTIVE

BR-01: Unqualified Guarantees & Absolute Claims
BR-02: Current Pricing Standard ($49/mo vs legacy $29)
BR-03: Unsubstantiated Superlatives & Puffery
BR-04: High-Pressure Sales Language & FOMO CTAs
BR-05: Forward-Looking & Security Disclaimers
BR-06: WCAG 2.1 AA Visual Contrast & Inclusivity

Verified Corporate Facts

4 VERIFIED

AF-01: TRVST Starter is $49/mo billed annually.
AF-02: Certified SOC2 Type II and GDPR compliant.
AF-03: 99.95% target enterprise uptime SLA.
AF-04: Conceived & engineered by Team Spider.

Source-audited against finance and legal records.

Terminology Dictionary

6 GUARDRAILS

~~"100% risk-free"~~ → mitigate compliance risk
~~"zero errors forever"~~ → automated validation
~~"world's greatest"~~ → industry-leading benchmark
~~"cheap"~~ → cost-effective & high-ROI
~~"obliterate competitor"~~ → outperform workflows

Visual & Accessibility Standards (VS-01 & VS-02):

Enforces verified hex palettes (Navy #0c192c, Teal #00bfa5, Amber #f59e0b, Coral #f43f5e), 24px logo margin clearance, and minimum 4.5:1 text-to-background contrast.

07 / EXPLAINABLE FEEDBACK & FIXES

Severity Gating & 1-Click Inline Rewrites

Preserving creative intent: Every flag provides clear evidence, regulatory rationale, and an editable replacement.

Critical Flag (Publication Blocked)

SEVERITY 1

Flagged: "We guarantee 100% risk-free compliance with zero errors forever."

Citation: BR-01 (Unqualified Guarantees). Software cannot legally promise zero errors without safe-harbor qualification.

```
Suggested: "designed to mitigate compliance and brand risk with automated checks"
```

Moderate Flag (Fact Inconsistency)

SEVERITY 2

Flagged: "Upgrade to our new Enterprise Plan for only $29/mo!"

Citation: BR-02 (Current Pricing Standard). Legacy promotional rate of $29/mo was retired in Q2 2026.

```
Suggested: "$49/month billed annually"
```

Minor Flag (Tone & Superlatives)

SEVERITY 3

Flagged: "Our unmatched AI is the world's greatest protection."

Citation: BR-03 (Superlatives & Puffery). Enterprise voice guidelines require objective outcome-based framing.

```
Suggested: "industry-leading, benchmarked governance platform"
```

Dynamic 1-Click Substitution Engine: Clicking "Apply Rewrite" performs an atomic substring swap inside the active copy editor, triggers an immediate re-evaluation, updates the brand alignment dial, and logs the action to the audit trail.

Human-in-the-Loop Governance Loop

Ensuring automated intelligence never circumvents human accountability or organizational decision-making.

Role-Based Decision State Machine

Draft → In Review	Triggered upon content screening request
Changes Requested	Reviewer sends back with mandatory feedback notes
Escalated to Legal	Critical claims routed specifically to compliance officers
Approved for Publishing	Formal sign-off granted by authorized approver

Immutable Governance Audit Trail

Every decision, rewrite application, dismissal, and escalation is stamped with user identity, timestamp, and rationale:

```
[2026-10-24 18:30:10] Vritika Sheth (Brand Lead): Approved with Tone Revision
[2026-10-24 17:15:45] Radha Gautam (Legal): Critical Guarantee Overridden
[2026-10-24 16:40:12] Raghav Parashar (Creator): Applied 1-Click Compliant Rewrite
```

Active Persona Switcher in UI: Reviewers can toggle instantly between Creator, Brand Lead, Legal & Compliance, and Agency Freelancer personas to evaluate how actions and permissions dynamically adapt.

09 / VERIFICATION & DEMO READINESS

Three Built-In Evaluation Presets

Pre-configured benchmark scenarios accessible directly via toolbar buttons for instant, frictionless evaluation by judges.

Scenario 1: High-Risk Fintech Campaign

HIGH RISK

"Upgrade to our new Enterprise Plan for only $29/mo! We guarantee 100% risk-free compliance with zero errors forever. Act immediately..."

Expected Score: ~35 - 55 / 100 (Blocked)
Flags Triggered: 3 Critical (BR-01, TM-01, TM-02), 1 Moderate (BR-02, $29/mo pricing).
Fix Flow: 1-click rewrite replaces guarantees with compliant safe-harbor phrases.

Scenario 2: Stale Blog Announcement

ATTENTION NEEDED

"We are thrilled to announce our SOC1 compliance milestone! Our cheap starter package at $29 is designed to help teams save money..."

Expected Score: ~70 - 75 / 100 (Revisions Advised)
Flags Triggered: 1 Critical (SOC1 vs verified SOC2 Type II), 2 Moderate (TM-05 "cheap", stale $29 rate).
Fix Flow: Upgrades security claim to SOC2 Type II and repositions to "high-ROI".

Scenario 3: Fully Compliant Release

BRAND COMPLIANT

"Introducing TRVST AI Starter: available for $49/month billed annually. Backed by our 99.95% uptime SLA, SOC2 Type II compliance..."

Expected Score: 98 / 100 (Ready to Publish)
Flags Triggered: 0 Critical, 0 Moderate, 0 Minor.
Fix Flow: Immediate publication sign-off with green compliance badge.

JUDGE DEMO TIP: Click Scenario 1 → Run Review → Click "Apply Rewrite" → Watch the score recalculate live from 55 to compliant in 1 second!

Testing the Impact Hypotheses

Live analytics dashboard measuring risk reduction velocity and approval turnaround against Round 1 hypotheses.

RISK REDUCTION RATE
92%
Target: 90% reduction in brand and legal risk

APPROVAL VELOCITY
64%
Target: 60% faster approvals (4.2 hrs vs 2.5 days)

RISKS INTERCEPTED
19
High-risk claims intercepted pre-publication

ASSETS SCREENED
24
Screened across social, display, and email

Top Recurring Guideline Gaps

BR-01 (Absolute Guarantees & Risk)	6 flags
BR-04 (High-Pressure Sales CTAs)	5 flags
BR-02 (Outdated $29/mo Pricing Standard)	4 flags

Severity Breakdown Across Channels

21%	31%	48%
Critical	Moderate	Minor Tone

Modular Architecture & Global Edge Deployment

Built with zero single points of failure: deployed to GitHub Pages' global CDN with an optional FastAPI companion engine.

01. EXPERIENCE LAYER

Interactive Client

Single-Page Application powered by Tailwind CSS design system, responsive split-screen editor, animated SVG score gauges, and Chart.js.

02. SERVICES LAYER

FastAPI & Browser Core

Dual runtime flexibility: runs natively as an offline client-side SPA on GitHub Pages or via a Python 3.9+ FastAPI backend server.

03. INTELLIGENCE LAYER

Gemini 2.5 + Semantics

Direct multimodal Google AI Studio integration paired with an instant deterministic regex rule matcher for guaranteed offline reliability.

04. STORAGE LAYER

Vault & Audit Logs

Local storage and JSON persistence for brand rules, verified corporate facts, prohibited term dictionaries, and immutable human decision records.

Automated Pytest Test Suite:

All 6 comprehensive verification test suites (health checks, preset loading, high-risk flag detection, compliant passes, 1-click rewrite application, and analytics metrics) pass cleanly with 100% success rate.

Zero Serverless Cold Starts:

Deploying the fully self-contained engine directly to GitHub Pages ensures instant sub-millisecond page loads on GitHub's global edge without serverless container cold starts.

Team Spider — Cross-Functional Engineering

The six members responsible for product integration, AI checks, UX design, quality assurance, infrastructure, and presentation.

RP — Raghav Parashar
Team Lead & Full-Stack Architect

System design, FastAPI backend services, client-side SPA port, deployment pipeline, and end-to-end integration.

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VS — Vritika Sheth
AI / ML Engineer

Multimodal Gemini prompt architecture, Knowledge Vault retrieval rules, and visual contrast accessibility screening.

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TT — Tatvagna Tandel
Frontend & UI/UX Designer

Split-screen review interface, dark-navy theme system, animated score dial, and responsive mobile-first layouts.

tatvagna23@gmail.com

ST — Sujay Tudu
QA Specialist & Technical Writer

Test suite datasets, deterministic rule edge-case validation, documentation standards, and submission verification.

sujaykumar7345@gmail.com

TC — Tanay Chavan
Backend Developer

Data models, session handling, state machine transitions, audit trail persistence, and serverless optimization.

tanayc547@gmail.com

RG — Radha Gautam
Research & Presentation Lead

User persona analysis, legal risk research, hypothesis measurement framework, and presentation deck design.

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Product Delivery & Next Horizons

Delivered on schedule: fully working, pitch-ready intelligent brand governance with a clear technical roadmap.

DELIVERED IN ROUND 2 (MVP)

- Production UI on GitHub Pages
- Dual Gemini 2.5 + Semantic Engine
- 6 Brand Rules + 4 Facts + 6 Terms
- 1-Click Inline Rewrites
- 3 Pre-Configured Demo Scenarios
- Human Approval Loop & Audit Trail
- Real-time Chart.js Governance Dashboard

PLANNED FOR ROUND 3 (SCALE)

- Chrome Extension for in-browser review (Google Docs, Figma, CMS)
- Automated Figma & Canva plugin integration
- Multi-tenant SSO with Okta & Azure AD
- Automated regulatory law change monitors
- Custom vector embedding ingestion for enterprise PDFs

LIVE VERIFICATION LINKS

Live App:
https://noteforstudy1-tech.github.io/trvst-ai/

GitHub Repository:
https://github.com/noteforstudy1-tech/trvst-ai

1-Click Local Run: `./run.sh`

Engineering & Research Citation: This project was architected, engineered, and verified utilizing Google DeepMind Antigravity and advanced AI research for automated system synthesis, resilient dual-mode intelligence, and full-stack software delivery.